Two hundred alerts. One cause. Evidence attached.

For network operations teams — NOC leads, escalation engineers, and network architects — running WAN, campus, and provider-dependent environments where "is it us, or the carrier?" is a recurring, expensive question.

Correlix is RCA-first network observability: metrics, flows, logs, traps, and active path probes correlated into one ranked incident — root cause named, owner recommended, evidence attached, ticket filed.

HOW TO READ THIS DOCUMENT DEMONSTRATED runs today, shown live in any demo
ILLUSTRATIVE walkthrough scenario — timings are not measured benchmarks

01 What today's tools miss

Modern networks are not under-monitored — they are over-notified. Traditional platforms collect everything and conclude nothing; the hardest work, turning signals into a cause, is left to the engineer at 2 a.m. Three gaps do the damage, and they land hardest on the people this document is for:

GAP	WHAT HAPPENS	WHO FEELS IT
Alerts without causality	One fault produces two hundred notifications — each technically true, none of them the answer. The correlation happens in an engineer's head, under pressure, one console at a time.	The NOC lead staffing the alert queue; whoever owns MTTR.
Black-box AI verdicts	"AIOps" tools issue a verdict and hide the reasoning. A NOC cannot page a vendor or wake an executive on an unverifiable claim — so engineers re-verify everything by hand, and the automation saves nothing.	The escalation engineer whose name is on the page-out.
No responsibility seam	Tools stop measuring at the demarc. The provider says "no problem on our side," and both parties burn a night proving a negative.	The architect who owns the carrier relationship — and the bill.

02 Correlate. Corroborate. Prove.

Correlix ingests every telemetry plane into one correlation engine with a shared clock and a shared topology. The engine matches event patterns against built-in network fault-family signatures and statistical baselines, then does the thing no dashboard does: it corroborates independent streams into one ranked incident — root cause named, fault domain assigned, owner recommended — and files the ticket itself into ServiceNow or Jira with the evidence bundle attached. DEMONSTRATED

> The discipline that makes the verdict trustworthy: Correlix does not mark a root cause Confirmed until two independent telemetry streams — for example a control-plane event and a data-path measurement — degrade in the same activity window. Anything less is honestly labeled Suspected, with the missing evidence listed. Every claim in an incident clicks through to the raw signals behind it. The verdict shows its work.

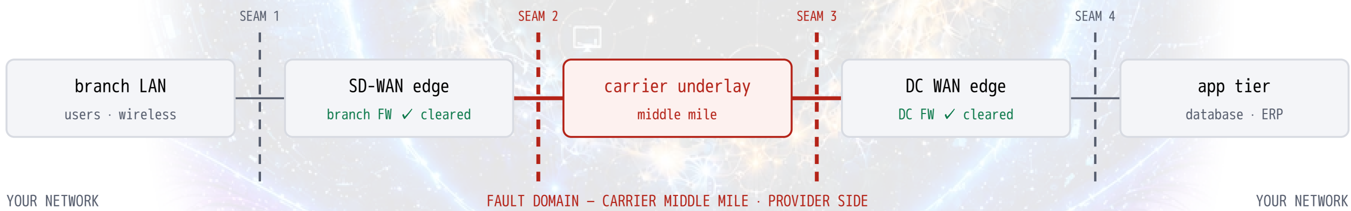
03 Use case 1 — "Is it us, or the carrier?" ILLUSTRATIVE

A user-to-database path crosses five domains, four seams, and two firewalls. Any of them could be the fault. Two independent streams pin the segment — with proof.

An enterprise runs branch users to data-center applications across a five-domain path: branch LAN → SD-WAN edge (with the branch firewall) → carrier underlay → data-center WAN edge (with the DC firewall) → application & database tier. Every handoff is a place the fault could hide — and a different team that could own it. On a quiet Tuesday night the path begins to degrade — not down, just *wrong*. The ERP front end stutters; nightly database replication crawls.

TIME	EVENT
22:04	Branch-to-DC replication slows; the path probe on that segment shows round-trip time climbing — 24 ms → 180 ms over ten minutes, with intermittent packet loss.
22:07	Retransmitting sessions pile up in the branch firewall's state table; its session-count and CPU alarms fire. In most NOCs, this is where the blame lands first.
22:09	The BGP session from the SD-WAN edge to the carrier flaps. Then again. Overlay tunnels re-key and fail over between underlays — the SD-WAN console still shows them "up".
22:15	Alert queues fill across every domain on the path — wireless clients, both firewalls, tunnel health, DC-edge interfaces, database slow-query alarms: 68 alerts across 14 devices and 5 domains.
22:19	Correlix confirms. Control-plane events (BGP to the carrier) and the independent path probe (RTT/loss on the same segment) degrade in the same activity window — while both firewalls' own telemetry shows the affected flows passing, zero drops, and the LAN- and DC-segment probes stay clean. Fault domain: carrier middle mile, provider side of the seam.

Without Correlix: five domains means five consoles and six owners. The app team blames the network; the NPM shows every interface up and green; the firewall team — guilty by default the moment its session alarms fired — spends the night exporting allow-logs to prove innocence; the SD-WAN dashboard says the tunnels are up because they failed over; and the carrier answers with the sentence every engineer knows by heart: *"We see no problem on our side."* By 23:30 there's a war room proving negatives, segment by segment.



Five domains, four seams — every handoff is a first-class object, so the fault domain names a *segment and a side*, not just a device. Corroboration: BGP flapping to the carrier (control plane) + RTT 24→180 ms with 4–9% loss (independent path probe) — same segment, same window. Exoneration works the same way: both firewalls are cleared by their own telemetry — the affected flows passing, zero drops — and the LAN- and DC-segment probes stay clean.

THE TRADITIONAL NIGHT

68 alerts · 14 devices · 5 domains · 6 owners

A war room assembling at 23:30 — the firewall guilty until proven innocent, every team proving a negative.

WITH CORRELIX

1 confirmed incident · fault domain: carrier middle mile · recommended owner: carrier

Both firewalls exonerated by their own telemetry. One ticket filed — evidence bundle attached. The carrier escalation opens with proof, not opinion.

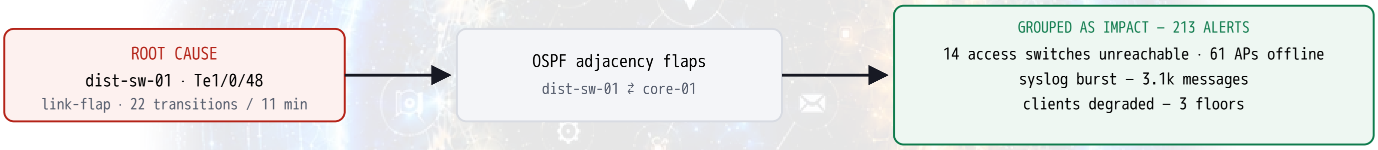
04 Use case 2 — 200 alerts, one failing uplink ILLUSTRATIVE

A degrading optic takes three floors with it. Correlix separates the cause from 213 symptoms.

Tuesday, 09:12, the campus full: on a distribution switch the optic on uplink `Te1/0/48` begins to fail — not dead, *flapping* (22 state transitions in eleven minutes). Each flap tears down the OSPF adjacency riding the link. Downstream, access switches and APs fall intermittently unreachable — and every one of them reports it. By 09:23: 214 device-level alerts, every one marked critical. The actual cause is alert #147 — a one-line interface warning buried between two hundred consequences.

```
09:12:04 CRIT ap-3f-17 lost contact with controller
09:12:09 CRIT sw-acc-3f-02 unreachable – ping timeout
09:12:11 WARN dist-sw-01 OSPF neighbor 10.20.0.1 down on Te1/0/48  ◀ the cause
09:12:12 CRIT ap-2f-04 lost contact with controller
09:12:15 CRIT sw-acc-2f-01 unreachable – ping timeout
09:12:21 CRIT voice-gw-02 station registrations dropping
+208 more in the next eleven minutes – every alert claiming equal urgency
```

With Correlix: the event pattern matches two built-in fault-family signatures — *link-flap* and *routing-adjacency-flap* — and the topology says everything alarming sits downstream of one interface. The causal graph pins `Te1/0/48` as the root; the other 213 alerts are grouped under it as impact — evidence of blast radius, not competing emergencies. DEMONSTRATED



Cause on the left, consequences grouped on the right. 213 alerts stop competing for attention the moment they are classified as impact — and close together when the optic is swapped.

THE INCIDENT CARD – INC-2481 · UPLINK LINK-FLAP – DIST-SW-01 · TE1/0/48

Confidence	Confirmed — 2 independent streams (interface state + routing adjacency events)
Fault domain	Inside — access/distribution uplink
Root cause	Degrading optic · 22 link transitions / 11 min
Impact	213 alerts grouped · 75 devices · 3 floors
Recommended owner	Campus network team
Recommended action	Replace optic on `Te1/0/48` · redundant uplink verified available
Evidence	214 signals · 4 sources · causal graph + evidence timeline attached · one ticket auto-filed

05 Will it fit your environment?

Deployment. Correlix is self-hosted: the full stack runs from a single Docker Compose install on one Linux host, behind one port. Your telemetry never leaves your network. Device onboarding is a subnet sweep — point discovery at your management ranges from the console and the inventory, topology, and collection start filling in. DEMONSTRATED

PLANE	INPUTS TODAY
Metrics	SNMP v2c/v3 (multi-vendor profiles), gNMI streaming, NETCONF
Events / logs	Syslog (per-vendor parsing incl. NGFW app/threat events), SNMP traps
Flows	sFlow, NetFlow, IPFIX — with application-identity enrichment
Active measurement	Path probes (STAMP-based RTT/jitter/loss), traceroute path discovery, per-WAN-interface SLA targets
Context	Auto-discovered topology; sites & source-of-truth import (CSV/JSON, NetBox connector)
Outputs	ServiceNow / Jira auto-ticketing with evidence bundle; Slack, PagerDuty, email, SMS notifications; REST API

Operational guardrails. Per-tenant isolation enforced at every storage layer; role-based access with SSO (OIDC/LDAP/TACACS+); credentials encrypted at rest in a built-in vault; every verdict carries its clickable evidence log. The optional AI assistant answers from your live data with citations — it is off by default, bounded, and never required for the RCA itself.

Proof maturity, stated honestly. Correlix has no public customer case studies yet — that is what the pilot below exists to produce, and we would rather show you than tell you. What is real today, and shown live in any demo: the correlation engine, fault-family signatures, seam corroboration (control-plane + independent path probe), alert-storm grouping, and the auto-ticketing loop all run against a physical multi-vendor lab with real fault injection — BGP instability, adjacency loss, link flaps, firewall drops — and ticket creation validated end-to-end against a ServiceNow-format instance. DEMONSTRATED

06 Why this works — and why you'd care

DESIGN DECISION	WHAT IT BUYS YOUR TEAM
Every plane, one platform	Metrics, flows, logs, traps, and probes share one clock and one topology. The cross-plane comparison that solves incidents happens automatically — no tool-hopping, no timeline reconstruction across three consoles.
Evidence, or it didn't happen	Confirmed requires two independent streams; Suspected says so out loud and lists what's missing. Your engineers stop re-verifying the automation — the re-verification is built in.
The responsibility seam, built in	Fault domains are modeled on both sides of the demarc. When the evidence points outward, the carrier escalation opens with proof attached — "no problem on our side" stops ending the conversation.
Closes the loop	One ticket per root cause, auto-filed with evidence. Incident time decomposition shows where the minutes went — detect, correlate, isolate, assign — so you manage what you can finally measure.

07 Next step — a 30-day pilot, scoped

Run Correlix on one domain of your network — a campus site, a WAN region, or a data-center pod. It discovers your devices, collects every plane, maps the topology, and correlates real incidents on your traffic and your provider seams. You judge the verdicts against what your engineers already know to be true. That is the test we want.

WEEK	MILESTONE
Week 1	Deploy (one Linux host, Docker), subnet discovery, topology mapped, telemetry planes green.
Week 2	Baselines established; path probes on the circuits that matter; ticketing connected (sandbox is fine).
Week 3	First evidenced incident — live if your network provides one, otherwise injected in a change window.
Week 4	Findings review against the scoreboard below; keep-or-kill decision with data.

Prerequisites: one Linux host, SNMP read credentials, syslog/flow export toward it, and optionally a ServiceNow/Jira sandbox. You keep: the discovered topology and inventory, the incident evidence bundles, and the scoreboard.

The pilot scoreboard — measured on your network, filled in together (no vendor-supplied numbers): incidents detected and their Confirmed/Suspected split · alerts collapsed per incident · time from first symptom to fault domain · owner-assignment accuracy judged by your engineers · duplicate tickets avoided · evidence completeness on escalations.

> Two hundred alerts. One cause. Evidence attached. — see it on your own network.

Scenario timings marked *Illustrative* are walkthroughs, not measured benchmarks.